Course Information

Course Number:	DAEN 459
Course Title:	Capstone Senior Design Planning
Section:	500
Time:	MW 10:20 AM-11:10 AM, F 1:50 PM-3:50 PM
Location:	MW ZACH 211, F ETB 1013
Credit Hours:	3 (2 hours of lecture, 3 hours of lab per week)

Instructor & TA/Grader

Instructor:	Alexander Abuabara	TA/Grader:	Nathan Marassa
Email:	abuabara@tamu.edu	Email:	nmarassa@tamu.edu
Zoom:	https://tamu.zoom.us/my/abuabara		
Office:	ETB 4013		
Office Hours:	MWF 11:30 AM-1 PM (or by appointment)		

Course Description

First in a two-course sequence for the capstone senior design experience; formation of a senior design team, visitation with the team sponsor, preparation of the groundwork for the project, preparation of the project charter and collection or acquisition of initial set of data; provision of instructions on different aspects of capstone design, including ethics, design constraints, applicable standards, project management, report writing specifications and requirements, and oral and visual presentations.

Learning Outcomes

At the end of this course, the student should be able to:

- Identify and articulate an open-ended design problem,
- Present proposal idea to a wide-ranging audience,
- Develop a project charter describing the steps to complete the senior design project, describing an open-ended problem into a concise data-engineering plan with measurable goals and constraints.

Course Prerequisites

Grade of C or better in DAEN 301

Course Corequisites

DAEN 400, DAEN 427, and DAEN 429

Special Course Designation

None

Course Expectations

As a data engineering student, you are expected to apply the skills and knowledge developed throughout your academic career toward the planning and development of a substantial, real-world data-centric project. These expectations mirror professional data engineering roles, preparing you for collaborative, cross-functional work. You are expected to:

- Contribute actively to your team with clear communication, accountability, and collaboration,
- Attend all meetings, meet deadlines, contribute to team decisions, and resolve conflicts professionally,
- Work with stakeholders to define and refine technical and business requirements,
- Develop a detailed project plan with timelines, data needs, and deliverables,
- Address ethical, privacy, and risk considerations,
- Identify, acquire, clean, and validate datasets, including the elaboration of data dictionaries and relevant metadata,
- Design a robust data architecture, including storage, pipelines, and processing frameworks,
- Document data flow, system architecture, and all planning and data engineering decisions with supportive analysis,
- Present project components clearly to both technical and non-technical audiences,
- Deliver a comprehensive design proposal with diagrams, models, and a clear roadmap,
- Track progress through milestone reviews and maintain an updated project board,
- Manage scope, time, and resources, adjusting plans as needed,
- Apply best practices in version control, reproducibility, and testing,
- Ensure compliance with industry standards, regulations, and school policies, and
- End the course with a feasible, well-scoped project ready for implementation.

Recommended Textbooks

- **Fundamentals of Data Engineering**. J. Reis, M. Housley (2022)
- **Designing Data-Intensive Applications**. M. Kleppmann (2017)
- **The Data Warehouse Toolkit**. R. Kimball, M. Ross (2013)

Grading Policy

- Graded items and weights:
  - **First draft of the project charter (25%)**: This will be a written report prepared by the team documenting the project objective and goals, the timeline, and the needs statement to complete the project charter at the end of the semester. This is due around week 4 of the semester.
  - **Second draft of project charter (25%)**: This will be a written report by the team documenting the progress made so far in creating the project charter. A well-defined projective objective statement will be stated based on the feedback from the review of the first draft. The report should document the data that was collected and the interactions with the sponsor of the project. The timeline should be suitably updated. This due around week 7-8 of the semester.
  - **Final project charter (30%)**: This is a final report that will be submitted by the team that develops the plan for the project and the steps needed to execute the project in the capstone senior design course (DAEN 460). The report is due at the end of the semester.
  - **Presentation of the project proposal (20%)**: The members of the team will present their project charter, the steps that were performed to collect all the required information, the analysis of the information, and the plan on how to execute the project in the follow-up course (DAEN 460). The presentations will occur during the last two weeks of the semester.
- In addition to the graded items listed above, other submissions will be evaluated on a **satisfactory/unsatisfactory** basis. Details of these requirements and milestones will be posted on the Canvas course page and may include review sessions as well as meetings with sponsors and advisors to discuss strategies.
- Grades will be calculated by totaling the points you earn and comparing them to the max. possible points.
- Grades assigned are A: 90%–100%, B: 80%–89.9%, C: 70%–79.9%, D: 60%–69.9% and F for less than 60%.

Graded Class Participation – None

Graded Attendance – None

Grades for Stacked Course – N/A

Grading Policy Changes – No changes after the second class; report issues promptly.

Late Work Policy

Late work will not be accepted except for approved university excused absences. Any disagreements regarding a grade received on any graded material must be discussed within one week of the return of the graded material. No grade will be changed beyond this limit. If you request a regrade, the entire submission will be regarded.

Course Assignments

This course focuses on developing a Project Charter for assessment. If time permits, students/teams will also outline the Project Management Packet. Guidelines are in the appendix; further instructions will be provided in class.

Attendance & Class Participation

Attendance is mandatory for sponsor meetings and presentation days. You are expected to attend all class lectures and labs except for university excused absences. The university rule regarding excused absences can be found at https://student-rules.tamu.edu/rule07.

Course Schedule

Tentative – Dates will not be changed without written notification to all students in the course (notice via email & Canvas).

Week	Topic	Deliverable / Due Date (TBC)
1	Course intro/overview, team formation, details about project sponsors **LAB**: Instructions on report requirements; report template	–
2	Project planning process, project charter components **LAB**: Start the work on project charter, develop draft project plan	–

Week	Topic	Deliverable / Due Date (TBC)
3	Interaction with sponsor/ planning a site visit **LAB**: Document sponsor requirements, draft the site visit plan statement	–
4	Developing the project charter **LAB**: Complete the first draft of project charter	**1st draft of project charter** (Fri Sep 19, 5 PM)
5-6	Engineering design, design constraints **LAB**: Develop relevant engineering design constraint requirements	–
7-8	Project management principles, revising project charter **LAB**: Complete the second draft of project charter	**2nd draft of project charter** (Fri Oct 17, 5 PM)
9	Ethics **LAB**: Draft the statement on relevant ethical issues for the project	–
10	Standards **LAB**: Complete the engineering standards section for the project	–
11-12	Site visit, preliminary data collection **LAB**: Rehearsal presentations (peer review & advisors/professors)	–
13	Completing the components of project charter **LAB**: Document details from site visit and data collection process	–
14-15	**Project proposal presentations (pitch to sponsors TBD)** **LAB**: Complete the project charter	**Final project charter** (sponsor-signed, Fri Dec 5, 5 PM)

Important Dates (https://registrar.tamu.edu/Academic-Calendar)

- Aug 25 First day of Fall semester classes
- Aug 29 Last day to add/drop courses for Fall semester
- **Sep 1 Labor day, no classes**
- **Oct 13-14 Fall break, no classes**
- Nov 19 Last day to drop courses with no penalty (Q-drop) or to withdraw (by 5 PM)
- **Nov 26 Reading day, no classes**
- **Nov 27-28 Thanksgiving holiday, no classes**
- Dec 8 Last day of Fall semester classes
- **Dec 9-10 Reading days, no classes**
- Dec 11-16 Final exams

Tips

(i) You all are important to me: include the course number in email subject so I can prioritize it. (ii) Every step matter and each build on the next: stay engaged, keep up, ask questions. (iii) Learning takes steady effort, start early, not last-minute cramming. (iv) Engineers turn data into meaningful information, practice here, get prepared. (v) Clear, well-organized work is as important as the content itself. (vi) Use others' ideas to build your own but always credit your sources: good scholarship means giving credit where it's due. (vii) No audio or video recording without permission; it's a conduct violation otherwise.

University Policies

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments. Please refer to Student Rule 7 in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Student Observances for Religious Holy Days

In accordance with Texas Education Code §51.911(b) and Texas A&M Student Rule 7: Attendance, students shall be excused from attending classes or other required activities, including examinations, for the observance of a religious holy day, including travel for that purpose. For more information about excused absences due to religious holy days, please visit the Faculty Affairs website (under Other University Guidelines).

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor. Please refer to Student Rule 7 in its entirety for information about makeup work, including definitions, and related documentation and timelines. Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" (Student Rule 7, Section 7.4.1).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" (Student Rule 7, Section 7.4.2).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See Student Rule 24.)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" (Section 20.1.2.3, Student Rule 20).

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Plagiarism Policy

According to the Texas A&M University Definitions of Academic Misconduct, plagiarism is the appropriation of another person's ideas, processes, results or words without giving appropriate credit (aggiehonor.tamu.edu). You should credit your use of anyone else's words, graphic images, or ideas using standard citation styles. AI text generators (ChatGPT, Google Bard, etc.) should not be used for any work for this class without explicit permission of the instructor and appropriate attribution. This includes, but is not limited to:

- *Creating or revising drafts*
- *Editing your work*
- *Reviewing a peer's work*

This excludes pre-existing software additions such as spelling and grammar checkers, which are acceptable.

Such use of AI text generators in this manner could be considered plagiarism and cheating according to Student Rule 20. More information may also be found at https://aggiehonor.tamu.edu or you may contact your instructor if you have questions.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact the Disability Resources office on your campus (resources listed below) Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

Disability Resources is in the Student Services Building or at (979) 845-1637 or visit disability.tamu.edu.

Title IX and Statement on Limits to Confidentiality

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.
With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office

if the employee experiences, observes, or becomes aware of an incident that meets the following conditions **(**see University Rule 08.01.01.M1):

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, a person who is subjected to the alleged conduct will be able to control how the report is handled, including whether to pursue a formal investigation. The University's goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

Students wishing to discuss concerns related to mental and/or physical health in a confidential setting are encouraged to make an appointment with University Health Services or download the TELUS Health Student Support app for 24/7 access to professional counseling in multiple languages. Walk-in services for urgent, non-emergency needs are available during normal business hours at University Health Services locations; call 979.458.4584 for details.

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the University's Title IX webpage.

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors influencing a student's academic success and overall wellbeing. Students are encouraged to engage in healthy self-care practices by utilizing the resources and services available through University Health Services. The TELUS Health Student Support app provides access to professional counseling in multiple languages anytime, anywhere by phone or chat, and the 988 Suicide & Crisis Lifeline offers 24-hour emergency support at 988 or 988lifeline.org.

Pregnancy Accommodations

Texas A&M provides reasonable accommodations to students due to pregnancy and/or related conditions, such as childbirth, recovery and lactation. Students should contact the University's Pregnancy Coordinator as soon as they become aware of the need for accommodation. Depending on the circumstances, accommodations could include extended time to complete assignments or exams, changes in course sequence, or modifications to the physical classroom environment. Texas A&M will also allow a voluntary leave of absence, ensure the availability of lactation space, and maintain grievance procedures to provide for the prompt and equitable resolution of complaints of sex discrimination. For information regarding pregnancy accommodations, email TIX.Pregnancy@tamu.edu.

Appendix – Initial guidelines

This capstone course serves as an integrative, high-impact educational experience that challenges students to integrate their accumulated knowledge and apply it to a complex, real-world problem. The capstone projects encourage students to adopt a hands-on approach by building solutions from the ground up rather than relying only on existing tools or templates. The goal is to improve essential skills such as project management, communication, and interdisciplinary analysis, thereby preparing students for professional or academic success. Figure 1 provides a broad overview of how data moves through a system, highlighting the sequential steps of data engineering and the critical underlying disciplines that ensure its success.

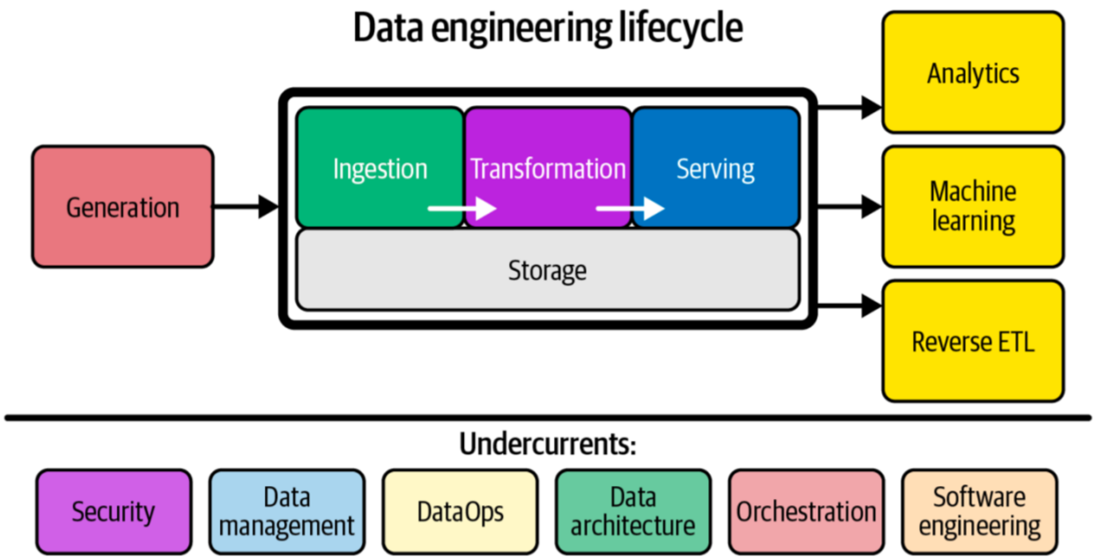

Figure 1 – Components and undercurrents of the data engineering lifecycle. Image extracted from ***Fundamentals of Data Engineering: Plan and Build Robust Data Systems****, J. Reis, M. Housley, 2022.*

Unlike *busy work* which often involves repetitive, low-cognitive tasks with limited connection to meaningful learning outcomes, the capstone project demands critical thinking, research, and collaboration. Here "reinventing the wheel" is not busy work but a deliberate method for gaining insights, building confidence, and preparing for real-world innovation. Drawing inspiration from Richard Feynman's quote, "What I cannot create, I do not understand", these are some of the great reasons to reinvent the wheel:

- Learn how wheels are made
- Teach others about wheels
- Learn about the inventors of wheels
- Be able to change wheels (or fix) when they break
- Build a better wheel (for some definition of better)
- Help someone in need of a very special wheel (maybe for a wheelchair?)
- Learn a tiny slice of what it means to build a larger system (such as a vehicle)

V-model

The V-model (Fig. 2) provides a structured framework ideal for planning the Data Engineering Capstone Senior Design courses. It guides students from defining project goals and requirements to detailed system design and implementation. This model promotes a detailed and careful planning, iterative development, and continuous verification, helping students understand the full lifecycle of a data engineering system, from concept and design to deployment and maintenance. Each phase on the left is mirrored by corresponding testing and validation on the right, ensuring alignment between design and outcome.

Within this framework, a Project Charter is typically the first formal document, authorizing the work, defining objectives, scope, stakeholders, and approach, and giving each team the authority to access necessary resources and tools. Once the charter is approved, the Project Management Packet is assembled, detailing schedules, risks, communications, and other operational plans, and ensuring execution stays consistent with the V-model. Like a movie, the charter is the *green light*; the packet is the *production plan*. Below you have more details about the Project Charter and the Project Management Pack.

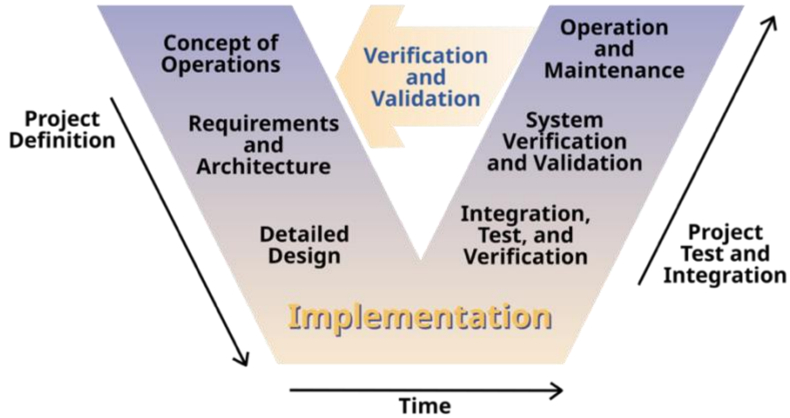

Figure 2 – Systems Engineering Process. Image extracted from ***Clarus: Concept of Operations, Federal Highway Administration (FHWA)****, 2005. (Publication No. FHWA-JPO-05-072, authors: L. Osborne, J. Brummond, R. Hart, M. Zarean, S. Conger.)*

1. Project Charter

The Project Charter is a concise, authoritative document that formally authorizes a project. In a two-semester capstone sequence, the final sponsor-signed charter is the *green light* to proceed to implementation in the second capstone semester. Endorsed by the project sponsor, it grants the project manager and team the authority to commit organizational resources and establishes a common understanding of the project's purpose, scope, and constraints. Treat the charter as both a contract and a living reference; if conditions change, revise it formally rather than drifting informally. These are important aspects that need to be addressed in the charter:

1. **Authority & accountability**: Sponsor sign-off clarifies who owns vision and acceptance.
2. **Unified vision**: Forces the team to articulate a shared understanding before deep technical work.
3. **Baseline for changes**: Scope or resource changes must be justified against the charter, avoiding unpleasant surprises.
4. **Early risks**: Listing assumptions and constraints reveals potential (ethical/legal/technical) risks early enough to plan.
5. **Communication tool**: Because it is short, executives and non-technical stakeholders will actually read it.

2. Project Management Packet

The Project Management (PM) Packet is a single living document that guides a data-engineering capstone team from idea approval through final hand-off. Think of it as the project's constitution that captures what you will build, how, by whom, and when so faculty, sponsors, and teammates always have one authoritative source of truth. Keep the packet lean (about 10 pages) and version-controlled, and revisit it whenever scope, risks, or constraints change. A well-maintained packet not only streamlines your work, but it also showcases professional project-management discipline to future employers. How it's used:

1. **Proposal phase**: Submitted with your charter to secure faculty and sponsor sign-off.
2. **Execution phase**: Updated after each sprint review; serves as the agenda for weekly stand-ups.
3. **Evaluation phase**: Forms the backbone of your final report and memo, proving objectives were met.

Key components of the PM package and their functions:

- **Work Description**: Plain-language summary of what the team will build or analyze.
- **Deliverables**: Defines concrete outputs (code, datasets, report, poster) to hand over.
- **Project Boundaries**: States what is in and out to guard against scope creep.
- **Acceptance Criteria**: Minimum conditions the sponsor must see to sign off.
- **Constraints**: Non-negotiable limits (tools, data privacy, semester length, etc.).
- **Assumptions**: Conditions taken as true (e.g. data access) that may later change.
- **Quality and Performance Metrics**: How accuracy, reliability, and efficiency will be judged.
- **Metrics & Verification Methods**: How each metric will be measured, tested, and logged.
- **Timeline and Milestones**: Semester checkpoints that trigger progress reviews and pivots.
- **Task Priority**: Ranks tasks so critical-path items are tackled first.
- **Risks & Risk Management Plans**: Lists top pitfalls and the team's mitigation / contingency moves.
- **Cost Estimates and Budgeting**: Forecasts hours, cloud credits, and purchases to stay on budget.
- **Resources**: People, software, hardware, and data the project will rely on.